

An Infinite Family of Integer Solutions to $z^2 + y^2z + 2x^3 + 1 = 0$

Abstract

We give a fully explicit infinite family of integer solutions to

$$z^2 + y^2z + 2x^3 + 1 = 0.$$

The proof is elementary and self-contained. The construction is based on a polynomial identity which makes $2X^3 - 1$ split as $(M - 2T^2)(M + 2T^2)$. Setting $x = -X$, $y = 2T$, and $z = M - 2T^2$ then gives the required equation.

1 Statement of the result

The goal is not to classify all integer solutions, but to prove unconditionally that infinitely many exist.

Theorem 1. *The Diophantine equation*

$$z^2 + y^2z + 2x^3 + 1 = 0$$

has infinitely many integer solutions. More precisely, for every odd integer $r \geq 3$, define

$$a = r^2 - 2, \quad n = \frac{a^2 - 1}{2}, \quad T = \frac{r(a - 1)}{2},$$

$$X = 2n^2 + a, \quad M = 4n^3 + 3an + 1.$$

Then

$$(x, y, z) = (-X, 2T, M - 2T^2)$$

is an integer solution of

$$z^2 + y^2z + 2x^3 + 1 = 0.$$

As r ranges over the odd integers $r \geq 3$, these solutions are pairwise distinct.

The same family can be written directly as a polynomial-parametric family:

Corollary 1. *For every odd integer $r \geq 3$, the triple*

$$\begin{aligned} x_r &= -\frac{r^8 - 8r^6 + 22r^4 - 22r^2 + 5}{2}, \\ y_r &= r^3 - 3r, \\ z_r &= \frac{r^{12} - 12r^{10} + 57r^8 - 134r^6 + 159r^4 - 84r^2 + 11}{2} \end{aligned}$$

is an integer solution of

$$z^2 + y^2z + 2x^3 + 1 = 0.$$

2 The auxiliary identity

We first isolate the identity behind the construction.

Lemma 1. *Let a and n be rational numbers satisfying*

$$n = \frac{a^2 - 1}{2}.$$

Define

$$X = 2n^2 + a, \quad M = 4n^3 + 3an + 1.$$

Then

$$M^2 - 2X^3 + 1 = \frac{(a-1)^4(a+2)^2}{4}.$$

Proof. We expand first without substituting the special value of n . Since

$$M = 4n^3 + 3an + 1, \quad X = 2n^2 + a,$$

we have

$$\begin{aligned} M^2 &= (4n^3 + 3an + 1)^2 \\ &= 16n^6 + 24an^4 + 8n^3 + 9a^2n^2 + 6an + 1, \end{aligned}$$

and

$$\begin{aligned} 2X^3 &= 2(2n^2 + a)^3 \\ &= 2(8n^6 + 12an^4 + 6a^2n^2 + a^3) \\ &= 16n^6 + 24an^4 + 12a^2n^2 + 2a^3. \end{aligned}$$

Therefore

$$M^2 - 2X^3 + 1 = 8n^3 - 3a^2n^2 + 6an - 2a^3 + 2. \tag{1}$$

Now substitute $n = (a^2 - 1)/2$. Then (1) becomes

$$\begin{aligned} &8 \left(\frac{a^2 - 1}{2} \right)^3 - 3a^2 \left(\frac{a^2 - 1}{2} \right)^2 + 6a \left(\frac{a^2 - 1}{2} \right) - 2a^3 + 2 \\ &= (a^2 - 1)^3 - \frac{3a^2(a^2 - 1)^2}{4} + 3a(a^2 - 1) - 2a^3 + 2. \end{aligned}$$

Multiplying by 4, the last expression is equal to

$$\begin{aligned} &4(a^2 - 1)^3 - 3a^2(a^2 - 1)^2 + 12a(a^2 - 1) - 8a^3 + 8 \\ &= a^6 - 6a^4 + 4a^3 + 9a^2 - 12a + 4 \\ &= (a - 1)^4(a + 2)^2. \end{aligned}$$

Dividing by 4 gives

$$M^2 - 2X^3 + 1 = \frac{(a-1)^4(a+2)^2}{4},$$

as claimed. □

3 Turning the identity into a solution

The next lemma is the conversion step. It shows how a factorization of $2X^3 - 1$ immediately gives a solution of the original equation.

Lemma 2. *Let $X, M, T \in \mathbb{Z}$ satisfy*

$$M^2 - 2X^3 + 1 = 4T^4.$$

Then

$$(x, y, z) = (-X, 2T, M - 2T^2)$$

is an integer solution of

$$z^2 + y^2z + 2x^3 + 1 = 0.$$

Proof. Define

$$x = -X, \quad y = 2T, \quad z = M - 2T^2.$$

Then

$$z + y^2 = M - 2T^2 + (2T)^2 = M + 2T^2.$$

Therefore

$$\begin{aligned} z(z + y^2) &= (M - 2T^2)(M + 2T^2) \\ &= M^2 - 4T^4. \end{aligned}$$

Using the assumed identity $M^2 - 2X^3 + 1 = 4T^4$, we get

$$M^2 - 4T^4 = 2X^3 - 1.$$

Thus

$$\begin{aligned} z^2 + y^2z + 2x^3 + 1 &= z(z + y^2) + 2(-X)^3 + 1 \\ &= (2X^3 - 1) - 2X^3 + 1 \\ &= 0. \end{aligned}$$

This proves the lemma. □

4 Proof of the infinite family

Proof of Theorem 1. Let $r \geq 3$ be odd. Define

$$a = r^2 - 2, \quad n = \frac{a^2 - 1}{2}, \quad T = \frac{r(a - 1)}{2},$$

$$X = 2n^2 + a, \quad M = 4n^3 + 3an + 1.$$

We first check that all quantities are integral. Since r is odd, r^2 is odd, so

$$a = r^2 - 2$$

is odd. Hence $a^2 - 1 = (a - 1)(a + 1)$ is divisible by 8, and in particular

$$n = \frac{a^2 - 1}{2} \in \mathbb{Z}.$$

Also $a - 1 = r^2 - 3$ is even, so

$$T = \frac{r(a - 1)}{2} \in \mathbb{Z}.$$

It follows immediately that $X, M \in \mathbb{Z}$.

By Lemma 1,

$$M^2 - 2X^3 + 1 = \frac{(a-1)^4(a+2)^2}{4}.$$

Because $a = r^2 - 2$, we have $a + 2 = r^2$. Hence

$$\begin{aligned} \frac{(a-1)^4(a+2)^2}{4} &= \frac{(a-1)^4 r^4}{4} \\ &= 4 \left(\frac{r(a-1)}{2} \right)^4 \\ &= 4T^4. \end{aligned}$$

Thus

$$M^2 - 2X^3 + 1 = 4T^4.$$

Lemma 2 now gives that

$$(x, y, z) = (-X, 2T, M - 2T^2)$$

is an integer solution of

$$z^2 + y^2 z + 2x^3 + 1 = 0.$$

It remains only to prove that the construction gives infinitely many distinct solutions. For $r \geq 3$, the number

$$a = r^2 - 2$$

is strictly increasing with r , and

$$n = \frac{a^2 - 1}{2}$$

is strictly increasing with a for $a \geq 7$. Therefore

$$X = 2n^2 + a$$

is strictly increasing as r ranges over the odd integers $r \geq 3$. Consequently the first coordinate

$$x = -X$$

assumes infinitely many distinct values. The corresponding triples are therefore pairwise distinct. $\square$

Proof of Corollary 1. Substituting $a = r^2 - 2$ into

$$n = \frac{a^2 - 1}{2}$$

yields

$$n = \frac{(r^2 - 2)^2 - 1}{2} = \frac{(r^2 - 3)(r^2 - 1)}{2}.$$

Then

$$\begin{aligned} X &= 2n^2 + a \\ &= 2 \left(\frac{(r^2 - 3)(r^2 - 1)}{2} \right)^2 + r^2 - 2 \\ &= \frac{r^8 - 8r^6 + 22r^4 - 22r^2 + 5}{2}. \end{aligned}$$

Also

$$2T = r(a - 1) = r(r^2 - 3) = r^3 - 3r.$$

Finally,

$$\begin{aligned} M - 2T^2 &= 4 \left(\frac{(r^2 - 3)(r^2 - 1)}{2} \right)^3 + 3(r^2 - 2) \left(\frac{(r^2 - 3)(r^2 - 1)}{2} \right) + 1 - 2 \left(\frac{r(r^2 - 3)}{2} \right)^2 \\ &= \frac{r^{12} - 12r^{10} + 57r^8 - 134r^6 + 159r^4 - 84r^2 + 11}{2}. \end{aligned}$$

These expressions are exactly the stated (x_r, y_r, z_r) . By Theorem 1, they solve the equation. $\square$

5 Examples

The following table lists several integer solutions produced by the construction. The final column records the value of

$$z^2 + y^2z + 2x^3 + 1,$$

which is 0 in each case.

r	x	y	z	$z^2 + y^2z + 2x^3 + 1$
3	-1159	18	55639	0
5	-139415	110	73611143	0
7	-2437679	322	5382395279	0
9	-19468879	702	121485805039	0
11	-100252919	1298	1419581333159	0

For instance, the first row gives

$$(x, y, z) = (-1159, 18, 55639).$$

A direct check may be written in the factorized form

$$55639(55639 + 18^2) = 3113725357 = 2 \cdot 1159^3 - 1.$$

Therefore

$$\begin{aligned} 55639^2 + 18^2 \cdot 55639 + 2(-1159)^3 + 1 &= (2 \cdot 1159^3 - 1) - 2 \cdot 1159^3 + 1 \\ &= 0. \end{aligned}$$

6 A sign symmetry

Since the original equation contains y only through y^2 , every solution (x, y, z) immediately gives another solution $(x, -y, z)$. Thus the displayed family also yields the companion solutions

$$(-X, -2T, M - 2T^2)$$

for the same odd parameters $r \geq 3$.

7 Conclusion

The elementary identity in Lemma 1 produces infinitely many factorizations

$$2X^3 - 1 = (M - 2T^2)(M + 2T^2).$$

The substitution

$$x = -X, \quad y = 2T, \quad z = M - 2T^2$$

then converts these factorizations into integer solutions of

$$z^2 + y^2z + 2x^3 + 1 = 0.$$

Because the parameter r can be chosen as any odd integer $r \geq 3$, and because the corresponding x -coordinates are pairwise distinct, the equation has infinitely many integer solutions.